

Item #	Subsystem / Component Description	Component Function	Potential Failure Mode	Potential Effect of Failure	Severity of Effect (1-10)	Potential Cause of Failure / Failure mechanism	Occurrence: Probability of Failure Cause (1-10)	Control (Ability to detect & mitigate)	Detection (1-10)	Risk assesment = RPN = Sev x Occur x Detect
				1 = Inconsequential 2 = Damage is mitigated, small repairs needed 4 = Decreased function / accuracy 6 = Loss of function and use 10 = Loss of function and extreme damage to F		1 = cannot think of any way 2 = designed to prevent with all known uses / customers 4 = Unlikely 6 = Rare, but have seen it happen 10 = Very likely to happen in a single test		1 = Good control will work every time 3 = Likely detection (slow rate+ good signals) 5 = Good detection, (moderate rate + signals) 8 = Bad detection (fast rate+ fast reactions) 10 = No chance to detect until too late		
1a	Bellows Weld	Seal between bellows and housing	Stress failure / cracking	Leakage, innacurate test results	5	Pressure exceeds weld limit	4	Inspect all the welds before use	6	120
1b	Bellows Weld	Seal between bellows and housing	Weld embrittlnent	Leakage, weld breaks	5	Improper weld	4	Inspect all the welds before use	4	80
1c	Bellows	Allows sealing and axial movement	Burst	Damage to internals, leakage, innacurate test results	8	Pressure exceeds bellows limit	3	Leakage or visible damage	4	96
1d	Bellows	Allows sealing and axial movement	fatigue failure	Damage to internals, leakage, innacurate test results	6	Conitnual use leads to fatigue failure	2	Failure after awhile so inspections will reveal fatigue	6	72
2a	Housing Center	Holds valve assy	Yield, stress failure	Damage to internals, lack of function	8	Stress concentration leads to failure	5	Leakage or visible damage	5	200
2b	Housing Outlet Thin Wall	Supports valve orifice	Deformation	Loss of function, innacurate test results	5	Pressure differential forces orifice into wall	5	Inspection and analysis during operation	9	225
2c	Housing Outlet Channel	Restricted diam channel from orifice	Deformation	Decreased function, small damage	6	Decreased diameter will be result in large pressure differential	6	Disturbances or change in outlet flow	8	288
2d	Housing Center Weld	Holds valve assy	Stress failure	Damage to internals, loss of function	8	Stress concentrations	5	Leakage or visible damage	4	160
2e	Housing inlet cone	Restricts flow to orifice	Yield, stress failure	Disturbs flow	2	Pressure differential load				0

2f	Housing Center walls	Space for orifice assembly	Yield, stress failure	Disturbs flow	5	Stress concentrations				0
3a	Valve stem	Links actuator and orifice	Buckling	Decreased function, innacurate test results	3	Axial load exceeds buckling limit	2	Inspection and analysis during operation	9	54
3b	Valve stem	Links actuator and orifice	Deformation	Decreased function, innacurate test results	3	Lateral load exceeds Factor of Safety	1	Inspection and analysis during operation	9	27
3c	Valve stem	Links actuator and orifice	fatigue failure	Decreased function, innacurate test results	3	Repeated actuation	3	Inspection and analysis during operation, replace stem after constanttttt repeated use	7	63
3d	Valve Orifice	Restricts gas flow	Deformation	Decreased function, innacurate test	5	Decreased diameter will be result in large pressure differential	5	Disturbances or change in outlet flow	6	150
3e	Orifice-Housing thin wall		Resistance to actuation	Decreased function, innacurate test results	3	Friction between components	4	Inspection and analysis during operation	6	72
3f	Orifice-Housing thin wall		Abrasive wear	Decreased function, innacurate test results	3	Friction between components	4	Inspection and analysis during operation	6	72
4a	Actuator radial weld	fixes actuator	Stress failure	Decreased function, Inaccurate test, small damage	6	Thermal cycling induces stress at weld toe	4	Inspect all welds before and during use	5	120
4b	Actuator-orifice subassembly	Actuation and flow control	Heat transfer lag	Innacurate test, heat transfer to fluid	3	Thermal mass difference between actuator and orifice delays response	6	Test temperature at base of actuator	4	72

4c	Actuator subassembly	Actuates valve stroke	thermal fatigue	Decreased function, Inaccurate test, small damage	3	Thermal deformation causes cyclic stress failures	5	Testing of temperature over operating temperature range	5	75
4d	Actuator fixturing bolt	Fixes actuator	fatigue failure	Loss of function, innacurate test results	3	Vibration loosening	3	Inspection and analysis during operation	4	36
4e	Actuator fixturing bolt		Loosening	May slightly decrease accuracy of test	2	Thermal expansion of bolt and members	6	Able to see if the bolt is loose and tighten it	2	24